United States Patent Application Publication	20250276893
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	TENTORI; Lorenzo et al.

MICROELECTROMECHANICAL SENSOR WITH IMPROVED EXTERNAL FLUIDIC COUPLING AND MANUFACTURING PROCESS THEREOF

Abstract

A microelectromechanical sensor includes a supporting body, a sensing structure including a measuring chamber and a sensitive element, the sensitive element being partially in the supporting body and facing the measuring chamber; and a cap coupled to the supporting body. The cap includes a buried cavity, inlet holes communicating with the environment external to the sensor and with the buried cavity, and coupling holes communicating with the measuring chamber and with the buried cavity. The inlet holes is in fluidic communication with the coupling holes by the buried cavity, and the inlet holes are offset with respect to the coupling holes.

Inventors: **TENTORI; Lorenzo (Verano Brianza, IT), AZPEITIA URQUIA; Mikel (Milano, IT)**

Applicant: **STMicroelectronics International N.V.** (Geneva, CH)

Family ID: **91023074**

Assignee: **STMicroelectronics International N.V. (Geneva, CH)**

Appl. No.: **19/056032**

Filed: **February 18, 2025**

Foreign Application Priority Data

IT	102024000004465	Feb. 29, 2024

Publication Classification

Int. Cl.: **B81B7/00** (20060101); **B81C1/00** (20060101)

U.S. Cl.:

CPC **B81B7/0061** (20130101); **B81C1/00309** (20130101); B81B2201/0264 (20130101); B81B2203/0127 (20130101); B81B2203/0315 (20130101); B81B2203/033 (20130101); B81B2203/0353 (20130101); B81C2201/0105 (20130101); B81C2201/013 (20130101); B81C2203/0109 (20130101)

Background/Summary

BACKGROUND
Technical Field
[0001] The present disclosure relates to a microelectromechanical sensor with improved external fluidic coupling and to the manufacturing process thereof.
Description of the Related Art
[0002] Environmental quantity sensors, such as MEMS (microelectromechanical) barometric pressure sensors, humidity sensors, gas and chemical detectors and sensors, and flow sensors, generally comprise a sensing element packaged in a case and connected to the external environment through one or more inlet holes in the case and internal fluidic paths, to receive the quantity to be measured. In MEMS pressure sensors, for example, the sensing element is often defined by a membrane that is deformable in response to variations in external pressure: in this case, a pressure signal may be read, in a capacitive or resistive manner, with respect to a/an (internal) reference pressure. Inlet holes and internal fluidic paths allow external pressure variations to propagate to the sensing element.
[0003] Typically, the inlet holes are provided in peripheral regions of the case, or in any case at a distance from the sensing element, which is in general positioned in a central region of the sensor; for example, the inlet holes may be organized in rows along the edges of the case. Such a positioning of the inlet holes is preferable as it reduces the risk that, during the formation of the same, the etching (e.g., chemical etching) steps damage the sensitive surfaces of the sensing element.
[0004] However, in the final steps of the manufacturing process of the sensor, and in particular in the steps of dicing the wafers and packaging, there is a risk that the inlet holes obstruct—even only partially—due, for example, to the migration of co-molding resins (a phenomenon known as “clogging”). Furthermore, again during the final steps of the manufacturing process, and in any case during the average life of the sensor, potentially harmful particles, such as grains of dust, may reach the sensing element of the sensor through the inlet holes and the internal fluidic paths. Contaminating particles may affect the functionalities of the sensor, for example by modifying the capacitances between conductive paths and/or creating drifts of the sensing elements.
[0005] To overcome these issues, a dedicated design of the inlet holes is often used, both in terms of position and in terms of number and dimensions (for example, diameters equal to 10 μm). However, there are constraints that may limit the effectiveness of measures of this kind. For example, reducing the number of inlet holes may not be compatible with the desirable redundancy for this type of sensors, while too small dimensions may even more easily cause obstruction of the inlet holes.
BRIEF SUMMARY
[0006] The present disclosure is directed to overcome or at least in part mitigate the disadvantages and limitations of the state of the art.
[0007] According to the present disclosure, a microelectromechanical sensor with external fluidic coupling and the manufacturing process thereof are provided.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0008] For a better understanding of the present disclosure, some embodiments are provided, by way of non-limiting example, with reference to the attached drawings, wherein:

[0009] FIG. **1** schematically shows, in a cross-section, a MEMS sensor in accordance with an embodiment of the present disclosure;

[0010] FIG. **2** schematically shows, in a top-plan view, the MEMS sensor of FIG. **1**;

[0011] FIG. **3** schematically shows in a cross-section a MEMS sensor in accordance with a different embodiment of the present disclosure;

[0012] FIGS. **4-8** schematically show, in top-plan views, different embodiments of a component of the MEMS sensor of FIG. **3**;

[0013] FIGS. **9***a*-**9***d* schematically show, along the same cross-section of FIG. **3**, consecutive steps of a manufacturing process of the MEMS sensor of FIG. **3** in accordance with an embodiment of the present disclosure; and

[0014] FIGS. **10***a*-**10***g* schematically show, along the same cross-section of FIG. **3**, consecutive steps of a manufacturing process of the MEMS sensor of FIG. **3** in accordance with a different embodiment of the present disclosure.

DETAILED DESCRIPTION

[0015] The following description refers to the arrangement shown in the drawings; consequently, expressions such as "above," "below," "upper," "lower," "top," "bottom," "right," "left" and the like relate to the attached Figures and are not to be interpreted in a limiting manner.

[0016] FIGS. **1** and **2** illustrate a portion of a microelectromechanical (MEMS) sensor with external fluidic coupling, for example a barometric pressure sensor (or simply, pressure sensor), in accordance with an embodiment of the present disclosure and indicated as a whole with the number **1**. The pressure sensor **1** comprises a supporting body **2**, containing semiconductor material (for example intrinsic and/or doped silicon), and a cap **3**, also of semiconductor material. The cap **3** overlies the supporting body **2** and has an internal surface **3***a*, arranged facing the supporting body **2**, and an external surface **3***b* opposite to the internal surface **3***a*. The cap **3** is also bonded to the supporting body **2** by a bonding ring **4**. A thickness of the bonding ring determines a distance between the cap **3** and the supporting body **2**. Considering a reference system of orthogonal axes X, Y, Z, the supporting body **2** and the cap **3** have a substantially planar shape parallel to the XY plane and are stacked in the direction of the Z axis.

[0017] The supporting body **2** accommodates a sensing structure **6** which occupies a substantially central position in the pressure sensor **1**. The sensing structure **6** is in communication with the environment external to the pressure sensor **1** through the cap **3**, as explained below. In practice, the supporting body **2**, the cap **3** and the bonding ring **4** delimit a volume which is accessible from the outside through the cap **3**, allowing an external pressure PEXT (for example the atmospheric pressure) and/or other physical parameters (such as, for example, humidity) to reach the supporting body **2**.

[0018] The sensing structure **6** comprises in detail: a reference chamber **7**, permanently sealed with respect to the external environment and maintained at a reference pressure (internal pressure, P.sub.0); a membrane **8** of semiconductor material, which delimits on one side the reference chamber **7**; and a measuring chamber **9**, extending above the membrane **8** on a side opposite to the reference chamber **7** and in fluidic coupling with the external environment. Even more in detail, the membrane **8** is interposed between the reference chamber **7** and the measuring chamber **9** and has a lower face **8***a*, arranged facing the reference chamber **7**, and an upper face **8***b*, arranged facing the measuring chamber **9**. The reference chamber **7** is a cavity buried in the supporting body **2**.

[0019] The membrane **8** is suspended on the reference chamber **7** so that the deformations of the membrane **8** may occur along the direction parallel to the Z axis and therefore perpendicularly to the supporting body **2**: as the external pressure PEXT varies with respect to the reference pressure P.sub.0 of the reference chamber **7**, i.e., as the pressure on the upper face **8***b* varies, the membrane **8** is free to deform and the deformation may be read by exploiting electrical phenomena. For example, the membrane **8** is provided with piezoelectric sensing structures—comprising implanted piezoresistances **7***a*—which respond to deformations of the same membrane **8**. In an alternative embodiment (not shown), the membrane **8**, that is conductive and suitably insulated from the supporting body **2**, may be capacitively coupled to reference electrodes placed at the base of the reference chamber **7**; in this case, an offset of the lower face **8***a* of the membrane **8** causes a variation in the capacitive coupling detectable at external terminals.

[0020] The pressure sensor **1** comprises a platform (also known as "decoupling mass") **20** which at least partially accommodates the sensing structure **6**. The platform **20** is suspended with respect to the supporting body **2** by flexures (not shown in FIGS. **1** and **2**) and is separated from the supporting body **2** by a separation region **25**. In a non-limiting embodiment, the separation region **25** is a trench which surrounds the platform **20** and has a substantially spiral shape in sections along planes parallel to the XY plane. A first portion **25***a* of the separation region **25** extends below and at least for the entire dimension of the platform **20**. Second portions **25***b* of the separation region **25** extend transversely to the first portion **25***a* (here parallel to the Z axis) and are communicating therewith. Ones of the second portions **25***b* are spaced from each other longitudinally along the first portion **25***a*. Conductive lines **9***a* for connecting the pressure sensor **1** extend, for example, above the platform **20**. The platform **20** improves the resistance of thermomechanical stresses, due for example to the packaging steps and/or the soldering steps of the pressure sensor **1**.

[0021] The cap **3** comprises, in detail, a plurality of inlet holes **5**, a buried rerouting cavity **10** and a plurality of coupling holes **15**.

[0022] The inlet holes **5** are arranged, for example, around the sensing structure **6** and extend along the Z axis from the external surface **3***b* of the cap **3** to the rerouting cavity **10**. The inlet holes **5** then couple the rerouting cavity **10** directly to the environment external to the pressure sensor **1**. In a non-limiting embodiment, the inlet holes **5** have a circular section and are organized in two rows parallel to the Y axis. The coupling holes **15** extend along the Z axis from the internal surface **3***a* of the cap **3** to the rerouting cavity **10** and communicate with the measuring chamber **9** of the sensing structure **6**. In a non-limiting embodiment, the coupling holes **15** have a circular section and are misaligned with respect to the inlet holes **5**. For example, the coupling holes **15** overlay the sensing structure **6** and are arranged in more internal positions of the pressure sensor **1**, along the X axis and/or the Y axis, with respect to the inlet holes **5**. For example, projections of the coupling holes **15** are surrounded by projections of the inlet holes **5** on the XY plane. In another non-limiting embodiment, the projections of the inlet holes **5** and a projection of the sensing structure **6** on the XY plane are disjointed, and the projections of the coupling holes **15** are overlapped with the projection of the sensing structure **6** on the XY plane.

[0023] The rerouting cavity **10** is buried in the cap **3** and puts the plurality of inlet holes **5** in communication with the plurality of coupling holes **15**. In detail, the rerouting cavity **10** extends at least partially over the sensing structure **6** and the platform **20**: for example, the rerouting cavity has a dimension along the X axis comprised between 50 μm and 1 mm and a dimension along the Y axis comprised between 50 μm and 1 mm. The rerouting cavity **10** also has a height H along the Z axis, comprised for example between 100 nm and 10 μm. In a non-limiting embodiment, the rerouting cavity **10**, moreover, extends parallel to the XY plane and protrudes laterally with respect to the inlet holes **5**.

[0024] Each inlet hole **5** puts the environment external to the pressure sensor **1** in communication with the rerouting cavity **10**. Each coupling hole **15** puts the rerouting cavity **10** in communication with the measuring chamber **9** of the sensing structure **6**. As already mentioned, moreover, the inlet holes **5** are misaligned with respect to the coupling holes **15**, so as not to have substantially straight and direct paths from the external environment to the measuring chamber **9**. In one embodiment, in particular, projections of the inlet holes **5** and the coupling holes **15** on the XY plane are disjoint. The measuring chamber **9** of the sensing structure **6** is therefore in communication with the environment external to the pressure sensor **1** through non-straight fluidic paths defined by the inlet holes **5**, the rerouting cavity **10** and the coupling holes **15**. Such a fluidic coupling provided by the rerouting cavity **10** is more robust against contamination of the sensing structure **6** by undesired particles owing to the absence of a direct straight path (along the Z axis) between the inlet holes **5** and the coupling holes **15**, i.e., between the environment external to the pressure sensor **1** and the measuring chamber **9**.

[0025] The rerouting cavity **10** of the pressure sensor **1** also comprises vertical traps that form collection reservoirs for the contaminating particles entering the pressure sensor **1** through the inlet holes **5**. In detail, first vertical traps **15***a* extend along the Z axis, towards the external surface **3***b* of the cap **3**, in positions corresponding to respective coupling holes **15**; second vertical traps **5***a* extend along the Z axis, towards the internal surface **3***a* of the cap **3**, in positions corresponding to respective inlet holes **5**. In practice, and as described below, each first vertical trap **15***a* is a continuation of a respective coupling hole **15** through and beyond the rerouting cavity **10**; each second vertical trap **5***a* is a continuation of a respective inlet hole **5** through and beyond the rerouting cavity **10**. In a non-limiting embodiment, the first and the second vertical traps **15***a*, **5***a* have, respectively, the same section along planes parallel to the XY plane of respective coupling holes **15***a* and respective inlet holes **5**. In another non-limiting embodiment, the rerouting cavity **10** is between the first and second traps **15***a*, **5***a* and the respective inlet holes **5** and the respective coupling holes **15**, and the first and second traps **15***a*, **5***a* are aligned with the respective inlet holes **5** and the respective coupling holes **15**.

[0026] The depth of the first and the second vertical traps **15***a*, **5***a* and the height H of the rerouting cavity are design parameters of the pressure sensor **1** with respect to reducing the probability that contaminating particles coming from the external environment reach the sensing structure **6**.

[0027] FIGS. **3** and **4** illustrate a pressure sensor in accordance with a different embodiment of the present disclosure and indicated as a whole with the number **50**. Elements of FIGS. **3** and **4** which correspond to elements of FIGS. **1** and **2** are illustrated with same reference numbers. Furthermore, the pressure sensor **50** is described below with reference to the differences with respect to the pressure sensor **1**.

[0028] The cap **3** of the pressure sensor **50** accommodates a rerouting cavity **60** and filtering structures **65**, accommodated into the rerouting cavity **60** and configured to limit the entrance of contaminating particles coming from the external environment. In detail, the filtering structures **65** comprise walls which extend through the rerouting cavity **60** and define obstacles for the passage of solid bodies between the plurality of inlet holes **5** and the plurality of coupling holes **15**, without the inlet holes **5** to be fluidically decoupled from the coupling holes **15**. The filtering structures **65** define, in the rerouting cavity **60**, inlet regions **61**, facing and directly communicating with respective inlet holes **5**, and a coupling region **62**, facing and directly communicating with the coupling holes **15**; each inlet region **61** communicates with the coupling region **62** by a respective filtering structure **65**.

[0029] In the pressure sensor **50**, the measuring chamber **9** of the sensing structure **6** is in communication with the external environment through tortuous fluidic paths defined by the filtering structures **65**. Some possible embodiments of the filtering structures **65** are described below, by way of example (therefore not to be understood in a limiting or exhaustive manner).

[0030] The rerouting cavity **60** of FIGS. **3** and **4** comprise a pair of filtering structures **65** that are distinct and identical and arranged so that, in the plan-view of FIG. **4**, the sensing structure **6** is comprised between the filtering structures **65**. Each filtering structure **65** is defined by a plurality of barrier elements **66** (three shown in FIGS. **3** and **4**) formed of the same material as the cap **3**, for example monocrystalline silicon. The barrier elements **66** of each filtering structure **65** extend in an elongated shape parallel to the Y axis and are offset to each other along the X axis by a pitch L. More in detail, the barrier elements **66** extend parallel to the Y axis with a dimension smaller than the dimension, along the Y axis, of the rerouting cavity **60**; furthermore, adjacent barrier elements **66** are defined starting from opposite walls (and parallel to the X axis) of the rerouting cavity **60**. Fluidic openings **66***a* are therefore present between the barrier elements **66** of a same filtering structure **65**. Fluidic openings **66***a* at external ends—with respect to the sensing structure **6**—of each filtering structure **65** are in communication with the respective inlet region **61**; fluidic openings **66***a* at internal ends of each filtering structure **65** are in communication with the coupling region **62**. In a non-limiting embodiment, the barrier elements **66** extend parallel to the Z axis for the entire height H of the rerouting cavity **60**.

[0031] The fluidic openings **66***a* of the filtering structures **65** allow the quantity to be measured to reach the sensing structure **6**; at the same time, the barrier elements **66** introduce a high resistance to the passage of contaminating particles. The dimensions of the openings **66***a*, the pitch L between the barrier elements **66** and the height H of the rerouting cavity **60** are in fact design parameters of the pressure sensor **50** with respect to the desired filtering capacity. For example, the openings **66***a* have a dimension along the Y axis<1 μm. In practice, the barrier elements **66** of the filtering structures **65** are integrated into the cap **3** and, more in particular, into the rerouting cavity **60**.

[0032] FIG. **5** shows a different embodiment of the filtering structures **65** of the rerouting cavity **60**. In particular, only the differences with respect to the embodiment of FIG. **4** are shown. More in detail, the barrier elements **66** extend, parallel to the Y axis, from side to side of the rerouting cavity **60**, each barrier element **66** being divided into portions **66′** separated from each other along the Y axis; respective fluidic openings **66***b* are present between adjacent portions **66′** of a same barrier element **66**. Fluidic openings **66***b* corresponding to adjacent barrier elements **66** are also offset parallel to the Y axis so as to avoid the presence of direct fluidic sections. The fluidic openings **66***b* of barrier elements **66** which are at the external ends—with respect to the sensing structure **6**—of each filtering structure **65** are in communication with the respective inlet region **61**; the fluidic openings **66***b* of barrier elements **66** which are at the internal ends of each filtering structure **65** are in communication with the coupling region **62**.

[0033] The filtering structures **65** of FIG. **5**, by creating a redundancy, reduce the probability that obstruction phenomena of the fluidic paths occur in the rerouting cavity **60**. The dimensions of the fluidic openings **66***b* are design parameters with respect to the desired filtering capacity. In a non-limiting embodiment, the fluidic openings **66***b* have dimensions along the Y axis that are equal to each other.

[0034] FIG. **6** shows another embodiment of the filtering structures **65** and, more generally, of the rerouting cavity **60**. In particular, only the differences with respect to the embodiment of FIG. **5** (or with respect to the embodiment of FIG. **4**) are shown. More in detail, the rerouting cavity **60** comprises trapping elements **67** which form collection reservoirs for the contaminating particles entering the pressure sensor **50** through the inlet holes **5** and/or for the contaminating particles present in the fluidic paths. The trapping elements **67** communicate directly with the rerouting cavity **60** with respect to which they are extensions parallel to the XY plane (“in-plane” trapping elements) and, optionally, also in the direction of the Z axis (“out-of-plane” trapping elements), as shown in FIG. **6**. In a non-limiting embodiment, the trapping elements **67** are defined both between the barrier elements **66** and directly in communication with a respective inlet region **61**. In practice, the trapping elements **67** are formed simultaneously with the rerouting cavity **60** and/or the filtering structures **65**.

[0035] FIG. **7** shows a further embodiment of the filtering structures **65** of the rerouting cavity **60**. In particular, only the differences with respect to the embodiments of FIGS. **4-6** are shown. More in detail, the filtering structures **65** comprise trenches which develop, in the XY plane, according to a serpentine profile (i.e., with folded arms) similar to that defined by the barrier elements **66** of FIG. **4** and with a depth equal to height H of the rerouting cavity **60**. The filtering structures **65** are in communication, on one side, with the respective inlet region **61** and, on the opposite side (in the direction of the X axis), with the coupling region **62**, through fluidic openings **66***c*. The filtering structures **65** may also comprise trapping elements **67** along the fluidic path, for example "in-plane" trapping elements, also defined according to the profile shown in FIG. **6**. In the embodiment of FIG. **7**, the serpentine profile of the filtering structures **65**—in terms of overall length and passage section—determines its filtering capacity.

[0036] Finally, with reference to FIG. **8**, in one embodiment of the pressure sensor **50**, the inlet holes **5** communicate with respective inlet regions **61** of the rerouting cavity **60**. The inlet regions **61** are portions of the rerouting cavity **60** defined by cells fluidically communicating with a coupling region **62** through respective fluidic openings **66***d;* each fluidic opening **66***d* creates a fluidic path between a respective inlet hole **5** and the plurality of coupling holes **15**. The fluidic openings **66***d* may be defined by straight sections whose section determines the filtering capacity (for example, a dimension along the Y axis<1 μm); in an embodiment not shown, each fluidic opening **66***d* is defined by a section having folded arms. The filtering structures **65** of FIG. **8** also comprise trapping elements **67**, for example defined by further cells fluidically communicating each with a respective inlet region **61** and a respective inlet hole **5**; the fluidic openings **66***d* show a high resistance to the passage of contaminating particles, while the trapping elements **67** show a high capacity to retain them.

[0037] Ultimately, by forming a rerouting cavity buried in the cap, the MEMS pressure sensor of the present disclosure has advantages in terms of design flexibility of the fluidic coupling of the sensing structure with the external environment. In particular, the sensor of the present disclosure is free of through holes that may form direct straight paths through the cap: the decentralization between inlet holes (communicating with the external environment) and coupling holes (communicating with the sensing structure) allows in fact the inlet holes to be defined in more central positions of the sensor and therefore the probability of obstruction phenomena to be significantly reduced during the packaging steps. Furthermore, filtering structures may be directly integrated into the rerouting cavity and therefore the risk of contaminations from the external environment may be reduced, without making modifications to the sensing structure. In particular, three-dimensional filtering structures may be created with shapes and dimensions as desired, obtaining filtering capacities that are high (<1 μm) and independent of the positioning and dimensions of the inlet holes. The filtering structures may finally cooperate with trapping elements of the rerouting cavity, i.e., collection reservoirs for contaminating particles. The contaminating particles coming from the external environment, therefore, tend to accumulate in dedicated positions and are further hindered in a possible route towards the sensing structure. In the pressure sensor, the probability that contaminating particles deposit in proximity to the membrane of the sensing structure is therefore significantly reduced, limiting variations in the electromechanical parameters during the life of the sensor. The pressure sensor of the present disclosure ultimately has more reliable measuring performance and repeatability.

[0038] The rerouting cavity of the present disclosure may be formed during the manufacturing process of the pressure sensor cap, as briefly shown in the embodiments of FIGS. **9***a*-**9***d* and FIGS. **10***a*-**10***g*. Hereinafter, reference is made to the rerouting cavity **60**, i.e., comprising filtering structures **65**, according to any of the embodiments of FIGS. **4**-**8**, but same manufacturing considerations also apply to the rerouting cavity **10** of FIG. **1**.

[0039] The manufacturing process of the rerouting cavity **60** according to the possible embodiments described below is independent of the manufacturing process of a sensor wafer **200**, of semiconductor material, intended to comprise the supporting body **2** and the respective components. Furthermore, starting from a cap wafer **300**, for example of monocrystalline silicon and intended to comprise the cap **3**, the variants described below share the same order of execution of the steps of: defining the filtering structures **65** (if any); forming the plurality of coupling holes **15**; subjecting to wafer flipping the cap wafer **300**; bonding to the sensor wafer **200**; and forming the plurality of inlet holes **5**.

[0040] With reference to FIGS. **9***a*-**9***d*, the rerouting cavity **60** and the integrated filtering structures **65** are created in the cap wafer **300**, according to one embodiment of the manufacturing process of the present disclosure.

[0041] In detail, the rerouting cavity **60** is created (FIG. **9***a*) by not-shown etching, epitaxial growth and annealing steps: in the etching step (for example an SF.sub.6 plasma etching) the filtering structures **65** and, in regions corresponding to the inlet regions **61** and to the coupling region **62**, structures separated by trenches (for example walls or pillars) may be defined; during the epitaxial growth, the trenches close without being filled; and during the annealing step (for example, an H.sub.2 annealing step) the silicon redistributes forming the inlet regions **61** and the coupling region **62** of the rerouting cavity **60**. Following the annealing step, the filtering structures **65** resulting from the etching step have dimensions determined by design preferences. During the processing steps leading to the cap wafer **300** of FIG. **9***a*, the cap wafer **300** has an orientation (along the Z axis) that is flipped with respect to the final orientation of the cap **3** of FIG. **3**, so that the internal surface **3***a* overlies the external surface **3***b*.

[0042] Therefore, referring to FIG. **9***b*, exploiting a special mask (not shown) applied to the internal surface **3***a*, the cap wafer **300** is selectively etched (for example by anisotropic etchings) at least to the rerouting cavity **60** so as to define the plurality of coupling holes **15**. The first vertical traps **15***a* may also be formed, with the same mask, as a continuation of the aforementioned selective etching of the internal surface **3***a* beyond the rerouting cavity **60**, for example with an anisotropic time-etching.

[0043] Successively, the cap wafer **300** is subject to wafer flipping assuming the final orientation of the cap **3**, as shown in FIG. **9***c*. The suspended platform **20**, the sensing structure **6** and the conductive lines **9***a* are formed in the sensor wafer **200**, separately. The cap wafer **300** is then bonded to the sensor wafer **200**, by the bonding ring **4** (step known as “wafer-to-wafer bonding”—FIG. **9***d*); such bonding is referred to as composite wafer **350** and it is understood that it is performed along the direction of the Z axis (the composite wafer **350** is shown in a same cross-section of the pressure sensor **50** of FIG. **3**). Known grinding and trimming processes of the composite wafer **350** and, in particular, of the external surface **3***b* of the cap wafer **300** then follow.

[0044] Finally, exploiting a special mask (not shown) applied to the external surface **3***b*, the cap wafer **300** is selectively etched (for example by anisotropic etchings) at least to the rerouting cavity **60** so as to define the plurality of inlet holes **5** and thus putting the sensing structure **6** in fluidic communication with the external environment. The second vertical traps **5***a* may also be formed, with the same mask, as a continuation of the aforementioned selective etching of the external surface **3***b* beyond the rerouting cavity **60**, for example with an anisotropic time-etching.

[0045] The final steps of the manufacturing process then follow, including the definition of external contacts and pads useful for extracting signals and the dicing operations of the composite wafer **350**, thereby obtaining the pressure sensor **50** of FIG. **3**.

[0046] It is understood that the pressure sensor **1** of FIG. **1** may be obtained by following the same steps of the manufacturing process as in FIGS. **9***a*-**9***d* and described above, except for the etching step leading to the definition of the cap wafer **300** of FIG. **9***a* (i.e., the filtering structures **65** are absent).

[0047] With reference to FIGS. **10***a*-**10***g*, according to a different embodiment of the manufacturing process of the present disclosure, the rerouting cavity **60** and the integrated filtering structures **65** are created in the cap wafer **300**.

[0048] In detail, the cap wafer **300** initially comprises a substrate **301**, for example of monocrystalline silicon, which has a front side **301***a* and a back side which defines the external surface **3***b* of the cap **3**. The substrate **301** is subject to an oxidation process (FIG. **10***a*) and is covered at least on the front side **301***a* by a sacrificial layer **302**, for example of silicon oxide; in a non-limiting embodiment, the sacrificial layer **302** coats the substrate **301** in its entirety. The sacrificial layer **302** is formed with a thickness coincident with the height H of the rerouting cavity **60** to be formed.

[0049] Successively, the sacrificial layer **302** is selectively etched and defined (FIG. **10***b*) on the front side **301***a* of the substrate **301**. In particular, first trenches **305** are opened in the sacrificial layer **302** along the profile of the filtering structures **65** to be formed; second trenches **305***a* are opened in the sacrificial layer **302** in positions corresponding to the coupling holes **15**, and therefore to the first vertical traps **15***a* where present in the cap **3**; the first and the second trenches **305**, **305***a* leave respective portions of the substrate **301** exposed. The sacrificial layer **302** is removed on the remaining front side **301***a* of the substrate **301** while it is maintained in the corresponding position and for the entire dimension of the rerouting cavity **60** to be formed (FIGS. **10***a*-**10***f* show the same cross-section as FIG. **3**).

[0050] A structural layer **303** is then formed, for example again of silicon, in contact with the substrate **301** and with the sacrificial layer **302** (FIG. **10***c*). The structural layer **303** may be grown in an epitaxial reactor starting from the semiconductor material of the substrate **301** and from a seed layer covering the sacrificial layer **302** (not shown for simplicity); the structural layer **303** therefore fills the first and the second trenches **305**, **305***a*. In detail, the structural layer **303** which is formed in contact with the substrate **301** is continuous therewith and identical from the point of view of the crystallographic shape. The structural layer **303** formed in the first and second trenches **305**, **305***a* and thereabove is therefore, for example, of monocrystalline silicon. The structural layer **303** which is formed in contact with the sacrificial layer **302** is instead, for example, of polycrystalline silicon. During the step of FIG. **10***c*, the material that fills the first trenches **305** forms the filtering structures **65**. The substrate **301** and the structural layer **303** together thus form the cap wafer **300** (flipped with respect to the final orientation of the cap **3** of FIG. **3**, so that the internal surface **3***a* overlies the external surface **3***b*). A free face of the structural layer **303** defines the internal surface **3***a* of the cap wafer **300**.

[0051] Therefore, referring to FIG. **10***d*, exploiting a special mask (not shown) applied to the internal surface **3***a*, the structural layer **303** is selectively etched (for example by dry etching) at least to the substrate **301** so as to define the plurality of coupling holes **15**. The first vertical traps **15***a* may also be formed, with the same mask, as a continuation of the aforementioned selective etching of the internal surface **3***a* in the substrate **301**, for example with a time-etching. The selectivity of the etching is ensured by the difference in material of the structural layer **303** and the sacrificial layer **302**.

[0052] Subsequently, the remaining sacrificial layer **302** (comprised between the structural layer **303** and the substrate **301** and coating the substrate **301**) is removed, for example by a hydrofluoric acid selective etching, thus releasing the rerouting cavity **60** (FIG. **10***e*), and in particular the inlet regions **61** and the coupling region **62**.

[0053] Successively, the cap wafer **300** is subject to wafer flipping assuming the final orientation of the cap **3**, as shown in FIG. **10***f*. The sensor wafer **200** is processed, separately, in order to form the suspended platform **20**, the sensing structure **6** and the conductive lines **9***a*. The cap wafer **300** is then bonded to the sensor wafer **200**, by the bonding ring **4** (step known as “wafer-to-wafer bonding”—FIG. **10***g*); such bonding is referred to as composite wafer **350** and it is understood that it is performed along the direction of the Z axis. Known grinding and trimming processes of the composite wafer **350** and, in particular, of the external surface **3***b* of the cap wafer **300** then follow.

[0054] Finally, exploiting a special mask (not shown) applied to the external surface **3***b*, the cap wafer **300** is selectively etched (for example by anisotropic etchings) at least to the rerouting cavity **60** so as to define the plurality of inlet holes **5** and thus putting the sensing structure **6** in fluidic communication with the external environment. The second vertical traps **5***a* may also be formed, with the same mask, as a continuation of the aforementioned selective etching of the external surface **3***b* beyond the rerouting cavity **60**, for example with an anisotropic time-etching.

[0055] The final steps of the manufacturing process then follow, including the definition of external contacts and pads useful for extracting signals and the dicing operations of the composite wafer **350**, thereby obtaining the pressure sensor **50** of FIG. **3**.

[0056] It is understood that the pressure sensor **1** of FIG. **1** may be obtained by following the same steps of the manufacturing process as in FIGS. **10***a*-**10***g*, except for the etching step leading to the definition of the sacrificial layer **302** of FIG. **10***b* (i.e., the first trenches **305** are absent).

[0057] It is also understood that the filtering structures **65** similar to those of FIGS. **4**-**6** and which implement comparable filtering capacities, may be formed in a manner complementary to what has been shown.

[0058] The manufacturing processes described reduce the possibility of damaging the sensing structure during the steps of opening the inlet holes as they are vertically offset with respect to the coupling holes: coupling holes and inlet holes of the sensor are in fact opened in distinct steps of the manufacturing process. Furthermore, as anticipated, the processing of the cap in order to create the structures described is independent of the processing of the supporting body and the elements that the supporting body accommodates. In particular, the manufacturing processes described are independent of the wafer-to-wafer bonding technology implemented (for example, thermal bonding or glass-frit bonding).

[0059] Finally, it is clear that modifications and variations may be made to what has been described and illustrated herein without thereby departing from the scope of the present disclosure.

[0060] For example, the sensing structure may comprise more than one piezoelectric membrane and/or elements sensitive to pressure variations of a nature different from what has been shown. More generally, the sensing structure may be different and may be capable to sensing different physical quantities, such as, for example, humidity.

[0061] Furthermore, the inlet holes and the coupling holes of the pressure sensor may be different from what has been shown, for example may have different geometry and/or be in different number and/or organized in a different geometric arrangement in the cap: the inlet holes may be for example organized in three rows surrounding the sensing structure on three distinct sides; or, the coupling holes may be arranged in more external positions of the pressure sensor with respect to the inlet holes; the vertical traps may also be in a different number than what has been shown or may be absent.

[0062] Still fulfilling the filtering function, the filtering structures may be different from what has been previously described: for example, the filtering structures may have geometries which in cross-section and in top-plan view are different from those shown in the attached Figures; or, the filtering structures may be defined by a combination of the embodiments shown. The rerouting cavity may also comprise (“in-plane” or “out-of-plane”) trapping elements and may be devoid of filtering structures.

[0063] The bonding ring useful for bonding the cap to the supporting body may be of any type (for example of metal material, with AlGe or AuAu compounds). Or, the cap may be defined on its surface arranged facing the supporting body so as to comprise cantilever elements which, once coated with a bonding layer (for example a germanium-aluminum layer), allow the supporting body to be bonded in dedicated positions.

[0064] The supporting body may also comprise structures for trapping contaminants coming from the external environment, such as for example trenches trenched into the supporting body and/or defined at least partially in the separation region, to further reduce the probability that such contaminants reach the sensing structure.

[0065] As regards the manufacturing process of the sensor of the present disclosure, it is understood that the step of bonding the cap wafer to the sensor wafer and the step of forming the plurality of inlet holes in the cap wafer may be reversed in the order of execution obtaining the same advantages described.

[0066] A microelectromechanical sensor (**1**; **50**) is summarized as including: a supporting body (**2**), containing semiconductor material; a sensing structure (**6**), including a measuring chamber (**9**) and a sensitive element (**8**), the sensitive element (**8**) being formed at least partially in the supporting body (**2**) and facing the measuring chamber (**9**); and a cap (**3**), of semiconductor material, coupled to the supporting body (**2**) and having an internal surface (**3***a*), arranged facing the supporting body (**2**), and an external surface (**3***b*), opposite to the internal surface (**3***a*) along a first direction (Z), wherein the cap (**3**) includes a buried cavity (**10**; **60**), a plurality of inlet holes (**5**), communicating with the environment external to the sensor and with the buried cavity (**10**; **60**), and a plurality of coupling holes (**15**), communicating with the measuring chamber (**9**) and with the buried cavity (**10**; **60**), wherein the plurality of inlet holes (**5**) is in fluidic communication with the plurality of coupling holes (**15**) by the buried cavity (**10**; **60**), and wherein the inlet holes (**5**) are offset with respect to the coupling holes (**15**).

[0067] The buried cavity (**10**; **60**) extends in the cap (**3**) perpendicular to the first direction (Z).

[0068] The inlet holes (**5**) extend parallel to the first direction (Z) from the external surface (**3***b*) at least to the buried cavity (**10**; **60**), and the coupling holes (**15**) extend parallel to the first direction (Z) from the internal surface (**3***a*) at least to the buried cavity (**10**; **60**).

[0069] The cap (**3**) further includes traps (**5***a*, **15***a*) extending parallel to the first direction (Z), beyond the buried cavity (**10**; **60**), in positions corresponding to respective inlet holes (**5**) and/or respective coupling holes (**15**).

[0070] The buried cavity (**60**) includes filtering structures (**65**) fluidically interposed between the inlet holes (**5**) and the coupling holes (**15**).

[0071] Each filtering structure (**65**) includes a plurality of barrier elements (**66**) arranged so as to create tortuous fluidic paths between respective inlet holes (**5**) and the plurality of coupling holes (**15**).

[0072] The barrier elements (**66**) have an elongated shape parallel to a second direction (Y), are offset to each other along a third direction (X) and have a height, parallel to the first direction (Z), equal to a height (H) of the buried cavity (**60**), the second and the third directions (Y, X) being perpendicular to each other and perpendicular to the first direction (Z), and each filtering structure (**65**) further includes fluidic openings (**66***a*; **66***b*), between adjacent barrier elements (**66**), offset at least parallel to the second direction (Y).

[0073] Each filtering structure (**65**) includes trenches which develop, in a plane (XY) perpendicular to the first direction (Z), according to folded arms, so as to create tortuous fluidic paths between respective inlet holes (**5**) and the plurality of coupling holes (**15**).

[0074] The buried cavity (**60**) further includes trapping elements (**67**) defined by trenches extending parallel to a plane (XY), perpendicular to the first direction (Z), and/or parallel to the first direction (Z).

[0075] The sensor (**1**; **50**) further includes a platform (**20**), suspended with respect to the supporting body (**2**) and accommodating the sensing structure (**6**) at least partially, and the sensing structure (**6**) is sensitive to pressure variations and includes a reference chamber (**7**) sealed at a reference pressure, and the sensitive element includes a membrane (**8**) interposed between the reference chamber (**7**) and the measuring chamber (**9**); the reference chamber (**7**) being a cavity buried in the supporting body (**2**), and the measuring chamber (**9**) being delimited by the supporting body (**2**) and the cap (**3**) and being in fluidic coupling with the environment external to the sensor through the cap (**3**).

[0076] A process for manufacturing a microelectromechanical sensor (**1**; **50**) is summarized as including: in a supporting body (**2**), containing semiconductor material, defining a sensing structure (**6**) including a surface sensitive element (**8**); in a cap (**3**), of semiconductor material and having an internal surface (**3***a*) and an external surface (**3***b*), opposite to the internal surface (**3***a*) along a first direction (Z), forming a buried cavity (**10**; **60**) and, on the internal surface (**3***a*), a plurality of coupling holes (**15**) communicating with the buried cavity (**10**; **60**); bonding the cap (**3**) to the supporting body (**2**) so as to define a measuring chamber (**9**) between the cap (**3**) and the supporting body (**2**), with the sensitive element (**8**) facing the measuring chamber (**9**), and so that

the coupling holes (**15**) communicate with the measuring chamber (**9**); and forming, on the external surface (**3***b*) of the cap (**3**), a plurality of inlet holes (**5**) communicating with the buried cavity (**10**; **60**), so that the plurality of inlet holes (**5**) is in fluidic communication with the plurality of coupling holes (**15**) by the buried cavity (**10**; **60**), wherein the inlet holes (**5**) are offset with respect to the coupling holes (**15**).
[0077] The manufacturing process includes forming, in the buried cavity (**60**), filtering structures (**65**) fluidically interposed between the inlet holes (**5**) and the coupling holes (**15**).
[0078] Forming the buried cavity (**10**; **60**) includes etching the cap (**3**), performing an epitaxial growth and an annealing in a reducing environment, before forming the plurality of coupling holes (**15**).
[0079] Forming the coupling holes (**15**) and the buried cavity (**10**; **60**) includes: forming a sacrificial layer (**302**) on a substrate (**301**) of semiconductor material; opening trenches (**305***a*) in the sacrificial layer (**302**) in positions corresponding to the coupling holes (**15**); forming a structural layer (**303**) of semiconductor material on the substrate (**301**) and on the sacrificial layer (**302**), filling the trenches (**305***a*) and creating the internal surface (**3***a*) of the cap (**3**); selectively etching the internal surface (**3***a*) to open the coupling holes (**15**); and removing the sacrificial layer (**302**) through the coupling holes (**15**), releasing the buried cavity (**10**; **60**).
[0080] The manufacturing process further includes opening further trenches (**305**) in the sacrificial layer (**302**) in positions corresponding to the filtering structures (**65**), and forming the structural layer (**303**) includes filling the further trenches (**305**) creating the filtering structures (**65**).
[0081] The various embodiments described above can be combined to provide further embodiments. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.
[0082] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. A microelectromechanical sensor comprising: a supporting body, containing semiconductor material; a sensing structure, comprising a measuring chamber and a sensitive element, the sensitive element at least partially in the supporting body and facing the measuring chamber; and a cap, of semiconductor material, coupled to the supporting body and having an internal surface facing the supporting body, and an external surface, opposite to the internal surface along a first direction, wherein the cap comprises a buried cavity, a plurality of inlet holes, communicating with the environment external to the sensor and with the buried cavity, and a plurality of coupling holes, communicating with the measuring chamber and with the buried cavity, wherein the plurality of inlet holes is in fluidic communication with the plurality of coupling holes by the buried cavity, and wherein the inlet holes are offset with respect to the coupling holes.
2. The sensor according to claim 1, wherein the buried cavity extends in the cap perpendicular to the first direction.
3. The sensor according to claim 1, wherein the inlet holes extend parallel to the first direction from the external surface at least to the buried cavity, and wherein the coupling holes extend parallel to the first direction from the internal surface at least to the buried cavity.
4. The sensor according to claim 1, wherein the cap further comprises traps extending substantially parallel to the first direction, beyond the buried cavity, in positions corresponding to respective inlet holes or respective coupling holes.
5. The sensor according to claim 1, wherein the buried cavity is between the traps and the

respective inlet holes and the respective coupling holes, and the traps are aligned with the respective inlet holes and the respective coupling holes.
6. The sensor according to claim 1, wherein the buried cavity comprises filtering structures fluidically interposed between the inlet holes and the coupling holes.
7. The sensor according to claim 6, wherein each filtering structure comprises a plurality of barriers, and tortuous fluidic paths between respective inlet holes and the plurality of coupling holes formed by the plurality of barriers.
8. The sensor according to claim 7, wherein the barriers have an elongated shape parallel to a second direction, are offset to each other along a third direction and have a height, parallel to the first direction, equal to a height of the buried cavity, the second and the third directions being perpendicular to each other and perpendicular to the first direction, and wherein each filtering structure further comprises fluidic openings, between adjacent barriers, offset at least parallel to the second direction.
9. The sensor according to claim 6, wherein each filtering structure includes trenches which develop, in a plane perpendicular to the first direction, according to folded arms, and tortuous fluidic paths between respective inlet holes and the plurality of coupling holes formed by the trenches.
10. The sensor according to claim 1, wherein the buried cavity further comprises traps defined by trenches extending parallel to a plane, perpendicular to the first direction.
11. The sensor according to claim 1, further comprising a platform, suspended with respect to the supporting body and accommodating the sensing structure at least partially, and wherein the sensing structure is sensitive to pressure variations and comprises a reference chamber sealed at a reference pressure, and wherein the sensitive element comprises a membrane interposed between the reference chamber and the measuring chamber; the reference chamber being a cavity buried in the supporting body, and the measuring chamber being delimited by the supporting body and the cap and being in fluidic coupling with the environment external to the sensor through the cap.
12. A process for manufacturing a microelectromechanical sensor comprising: in a supporting body, containing semiconductor material, defining a sensing structure comprising a surface sensitive element; in a cap, of semiconductor material and having an internal surface and an external surface, opposite to the internal surface along a first direction, forming a buried cavity and, on the internal surface, a plurality of coupling holes communicating with the buried cavity; bonding the cap to the supporting body so as to define a measuring chamber between the cap and the supporting body, with the sensitive element facing the measuring chamber, and the coupling holes communicating with the measuring chamber; and forming, on the external surface of the cap, a plurality of inlet holes communicating with the buried cavity, and the plurality of inlet holes is in fluidic communication with the plurality of coupling holes by the buried cavity, wherein the inlet holes are offset with respect to the coupling holes.
13. The manufacturing process according to claim 12, comprising forming, in the buried cavity, filtering structures fluidically interposed between the inlet holes and the coupling holes.
14. The manufacturing process according to claim 12, wherein forming the buried cavity comprises etching the cap, performing an epitaxial growth and an annealing in a reducing environment, before forming the plurality of coupling holes.
15. The manufacturing process according to claim 12, wherein forming the coupling holes and the buried cavity comprises: forming a sacrificial layer on a substrate of semiconductor material; opening trenches in the sacrificial layer in positions corresponding to the coupling holes; forming a structural layer of semiconductor material on the substrate and on the sacrificial layer, filling the trenches and creating the internal surface of the cap; opening the coupling holes by selectively etching the internal surface; and releasing the buried cavity by removing the sacrificial layer through the coupling holes.
16. The manufacturing process according to claim 15, further comprising opening further trenches

in the sacrificial layer in positions corresponding to the filtering structures, and wherein forming the structural layer comprises creating the filtering structures by filling the further trenches.
17. A device, comprising: a supporting body; a sensing structure including: a measuring chamber; and a sensitive element in the supporting body; and a cap coupled to the supporting body and having an internal surface facing the supporting body, and an external surface, opposite to the internal surface along a first direction, the cap includes: a buried cavity; a plurality of inlet holes coupled between an environment external and the buried cavity; and a plurality of coupling holes coupled between the measuring chamber and the buried cavity, the plurality of inlet holes are in fluidic communication with the plurality of coupling holes by the buried cavity, and the inlet holes are offset with respect to the coupling holes.
18. The device of claim 17, wherein the coupling holes are surrounded by the inlet holes.
19. The device of claim 17, wherein the cap includes a plurality of traps that are between the inlet holes.